

Science Exchange

Catalog Data Model

Products & Services

Internal Requirements Document — April 2026

1. Document Overview

This document defines the catalog data model for Science Exchange's marketplace, covering both products and services. It specifies universal attributes, category-specific product attributes, and service listing requirements to enable standardized search, filtering, and comparison across suppliers.

1.1 Purpose

Suppliers: Need a lightweight, easy-to-complete catalog upload process. Templates must map to data they already have in their systems. Required fields are limited to the absolute minimum needed to create a listing.

Buyers/Sponsors: Need standardized, filterable attributes to search and compare products and services across multiple suppliers.

1.2 Design Principles

Minimal supplier burden: Required fields are limited to 6–9 fields per listing type — data suppliers should already maintain.

Internal enrichment: Science Exchange enriches product data from public databases (PubChem, NCBI) and product name parsing rather than burdening suppliers.

Normalized search filters: Picklist fields enforce controlled vocabularies for consistent cross-supplier filtering.

Progressive data quality: Get suppliers onboarded with basic data first. Enrich and improve over time.

RFQ flexibility: Services without a fixed price operate under a Request for Quote flow, removing pricing as a barrier to listing.

1.3 Definitions

Required: Listing cannot be created without this field. Limited to data suppliers already have.

Recommended: Enhances search and buyer experience. Supplier is encouraged to provide but will not be blocked. May be enriched internally.

System: Auto-generated by the Science Exchange platform. Not part of the supplier upload.

SE Enriched: Science Exchange populates from public sources or product name parsing if supplier does not provide.

2. Product Categories

All products are classified into one of 11 SE categories. The category determines which category-specific attributes apply to a product.

Category	Description
Chemicals & Solvents	Lab chemicals, solvents, buffers, reagent-grade materials. Identified by CAS Number.

Molecular Biology Reagents	PCR primers & probes, gRNAs, Cas nucleases, RNAi reagents, enzymes, nucleotides.
Lab Supplies & Consumables	Plasticware, glassware, disposables, tubes, plates, pipette tips, safety supplies.
Antibodies	Primary and secondary antibodies. Search by target, host, reactivity, application.
Proteins & Peptides	Recombinant proteins, native proteins, cytokines, growth factors, peptides.
Kits & Assays	ELISA kits, cell isolation kits, detection assays, substrates, buffers, diluents.
Cell & Tissue Culture	Culture media, sera, supplements, cryopreservation reagents, matrices, substrates.
Biospecimens	Human and animal biological samples — primary cells, blood products, tissue.
Animal Models	Research animal models — mice, rats, zebrafish, genetically modified strains.
Equipment, Instruments & Parts	Lab instruments, accessories, replacement parts, consumable components.
Controls, Calibrators & Standards	QC controls, calibrators, reference standards, proficiency testing.

3. Universal Product Attributes

These fields apply to every product across all 11 categories. Required fields (highlighted in blue) represent the minimum a supplier must provide to create a listing.

Field	Definition / Examples	Priority	Format	Data Source
Product Identification				
Unique ID	System identifier. Populated from Supplier SKU.	Required	Text	System
Supplier SKU	Supplier's unique product identifier. e.g., AB-12345	Required	Text	Supplier
Manufacturer SKU	Original manufacturer part number. Include when supplier is a distributor.	Recommended	Text	Supplier
Manufacturer Name	Company that makes the product. Only populated when supplier ≠ manufacturer (distributor catalogs).	Conditional	Text	Supplier
UNSPSC Code	8-digit UN Standard Products and Services Code from the full UNSPSC commodity taxonomy.	Required	Text	Supplier / SE
Product Details				
Product Name	Descriptive name as shown in search. Preserved as-is from supplier.	Required	Text	Supplier
Product Description	Extended description. Preserved as-is from supplier.	Recommended	Text	Supplier
Category	SE product category. Determines which category-specific attributes apply. See Section 2 for full list of 11 categories.	Required	Picklist	SE Assigned

Type	"Product" for all physical goods.	Required	Picklist	SE Assigned
Product URL	Link to product page on manufacturer/supplier website.	Recommended	URL	Supplier
Product Image URL	Link to primary product image.	Recommended	URL	Supplier
Pricing & Packaging				
Price per Unit	Standard list price per unit. Never modified by SE.	Required	Numeric	Supplier
Currency	Currency code (e.g., USD, EUR, GBP). Preserved from supplier if provided; otherwise system-assigned on upload.	Recommended	Text	Supplier / System
Quantity	Number of units per package. e.g., 1, 6, 100	Required	Numeric	Supplier
Unit of Measure	How the product is sold. See Section 18 for full Product UOM picklist (37 values). Default to "Each" if no match.	Required	Picklist	Supplier
Size / Volume	Physical size or amount per unit. e.g., 500 mL, 100 µg, 96-well. Distinct from UOM — UOM is the selling unit, Size/Volume is how much product is in that unit.	Recommended	Text	Supplier / SE
Logistics				
Lead Time (Days)	Estimated business days to shipment.	Recommended	Numeric	Supplier
Ship From	Region product ships from. Values: US, EU, UK, China, Japan, India, Canada, Australia, Other. System-populated on upload.	System	Picklist	System
Site	Supplier site identifier. System-populated on upload.	System	Text	System
Remit To	Payment remittance entity. System-populated on upload.	System	Text	System
Documentation				
TDS URL	Link to Technical Data Sheet.	Recommended	URL	Supplier
SDS URL	Link to Safety Data Sheet.	Recommended	URL	Supplier
COA URL	Link to Certificate of Analysis.	Recommended	URL	Supplier
Additional Docs URL	Link to other supporting documentation.	Recommended	URL	Supplier

4. Category-Specific Product Attributes

Each product category defines additional fields beyond the universal set. Required fields are highlighted in blue. Only categories present in a given catalog need their columns included.

4.1 Chemicals & Solvents

2 additional Required fields (CAS Number and Purity). CAS Number enables auto-enrichment of molecular identifiers from PubChem.

Field	Definition / Picklist Values	Priority	Format	Data Source
CAS Number	Chemical Abstracts Service registry number. e.g., 7647-14-5	Required	Text	Supplier
Molecular Formula	Chemical formula. e.g., NaCl, C ₂ H ₅ OH	Recommended	Text	Supplier + SE
Molecular Weight	Molecular weight in g/mol. e.g., 58.44	Recommended	Numeric	Supplier + SE
Purity	Percentage or grade. e.g., ≥99%, ACS Grade	Required	Text	Supplier
Physical State	Values: Solid, Liquid, Gas, Powder, Crystal, Solution, Gel, Paste	Recommended	Picklist	Supplier
Hazard Classification	GHS categories (multi-select). Values: Flammable, Corrosive, Toxic, Oxidizer, Irritant, Environmental Hazard, Health Hazard, Explosive, Non-Hazardous	Recommended	Multi-Select	Supplier + SE
Storage Conditions	Values: Room Temperature, 2-8°C, -20°C, -80°C, Desiccated, Protected from Light	Recommended	Picklist	Supplier
Application	Intended research use.	Recommended	Text	Supplier

4.2 Molecular Biology Reagents

2 additional Required fields (Sub-Type and Target Gene/Region). Other fields are Recommended to avoid blocking suppliers whose products don't fit neatly into species or application categories.

Field	Definition / Picklist Values	Priority	Format	Data Source
Sub-Type	Values: Primer, Probe, siRNA, Enzyme, Nucleotide, gRNA, Cas Nuclease, shRNA, miRNA, Other	Required	Picklist	Supplier
Target Gene / Region	Gene, locus, or DNA/RNA region. e.g., BRCA1, TP53, GAPDH	Required	Text	Supplier
Target Species	Organism product is designed for. e.g., Human, Mouse, Rat	Recommended	Text	Supplier
Modification / Label	Chemical modifications or fluorescent labels. e.g., FAM, Biotin	Recommended	Text	Supplier

4.3 Lab Supplies & Consumables

2 additional Required fields (Sub-Type and Sterility). Material and Pack Size are Recommended because they're often embedded in the Product Name.

Field	Definition / Picklist Values	Priority	Format	Data Source
Sub-Type	Values: Tube, Plate, Pipette Tip, Filter, Cuvette, Flask, Glassware, Container, Syringe, Safety Supply, Other	Required	Picklist	Supplier
Material	Values: Polypropylene, Glass, Polystyrene, Polyethylene, PTFE, Nitrile, Latex, Stainless Steel, Other	Recommended	Picklist	Supplier
Sterility	Values: Sterile, Non-Sterile, Sterilizable	Required	Picklist	Supplier
Capacity / Volume	Functional capacity. e.g., 15 mL, 200 µL, 96-well	Recommended	Text	Supplier
Pack Size	Items per package. e.g., 100/pk, 500/cs	Recommended	Text	Supplier
Color	Product color.	Recommended	Text	Supplier
Compatibility	Equipment or chemical compatibility. e.g., Eppendorf, Autoclavable	Recommended	Text	Supplier
Temperature Range	Operating temp range. e.g., -80°C to 121°C	Recommended	Text	Supplier

4.4 Antibodies

5 additional Required fields. Antibody buyers search very specifically — Target, Host Species, Clonality, Reactivity, and Application are the minimum needed to identify relevance.

Field	Definition / Picklist Values	Priority	Format	Data Source
Target / Specificity	Protein or epitope. e.g., CD4, p53, Beta-Actin	Required	Text	Supplier
Host Species	Values: Mouse, Rabbit, Goat, Rat, Donkey, Chicken, Sheep, Other	Required	Picklist	Supplier
Clonality	Values: Monoclonal, Polyclonal, Recombinant Monoclonal	Required	Picklist	Supplier
Clone ID	Clone identifier. e.g., EPR3776, 2B11	Recommended	Text	Supplier
Reactivity	Species the antibody reacts with. Multi-select. e.g., Human, Mouse, Rat	Required	Multi-Select	Supplier

Application	Values: FC (Flow Cytometry), WB (Western Blot), IHC, IF, ELISA, IP, ChIP, ICC, Neutralization, Other	Required	Multi-Select	Supplier
Conjugate / Label	Values: Unconjugated, FITC, PE, APC, AF488, AF647, HRP, AP, Biotin, Cy3, Cy5, BV421, PE-Cy7, APC-Cy7, Other	Recommended	Picklist	Supplier
Isotype	e.g., IgG1, IgG2a, IgM	Recommended	Picklist	Supplier
Format	Values: Liquid, Lyophilized	Recommended	Picklist	Supplier
Storage Conditions	Values: 2-8°C, -20°C, -80°C	Recommended	Picklist	Supplier

4.5 Proteins & Peptides

2 additional Required fields (Source Organism and Form). Distinct from chemicals — buyers search by biological function and source organism.

Field	Definition / Picklist Values	Priority	Format	Data Source
Source Organism	Origin species or expression system. e.g., Human, Mouse, E. coli, HEK293, CHO	Required	Text	Supplier
Form	Values: Lyophilized, Liquid, Frozen	Required	Picklist	Supplier
Physical State	Values: Solid, Liquid, Powder, Solution	Recommended	Picklist	Supplier
Purity	e.g., >95% by SDS-PAGE	Recommended	Text	Supplier
Activity	Specific or enzymatic activity. e.g., ED50 < 1 ng/mL	Recommended	Text	Supplier

4.6 Kits & Assays

2 additional Required fields (Sub-Type and Application). Covers multi-component reagent packages, substrates, buffers, and assay components.

Field	Definition / Picklist Values	Priority	Format	Data Source
Sub-Type	Values: ELISA Kit, Extraction Kit, Detection Kit, Substrate, Blocking Buffer, Diluent, Stabilizer, Signal Enhancer, Lateral Flow, Other	Required	Picklist	Supplier
Application	Values: ELISA, Western Blot, IHC, Lateral Flow, Multiplex, Cell Isolation, Other	Required	Picklist	Supplier
Detection Method	Values: Chromogenic, Chemiluminescent, Fluorescent, Electrochemical	Recommended	Picklist	Supplier
Target Enzyme	Values: HRP, AP, Universal	Recommended	Picklist	Supplier

Physical State	Values: Liquid, Lyophilized, Powder	Recommended	Picklist	Supplier
Storage Conditions	Values: 2-8°C, -20°C, Room Temperature	Recommended	Picklist	Supplier

4.7 Cell & Tissue Culture

1 additional Required field (Sub-Type). Culture media, sera, supplements, growth factors, cryopreservation reagents, matrices.

Field	Definition / Picklist Values	Priority	Format	Data Source
Sub-Type	Values: Media, Supplement, Growth Factor, Serum, Cryopreservation Reagent, Matrix / Substrate, Dissociation Reagent, Other	Required	Picklist	Supplier
Species	Supported species. e.g., Human, Mouse, Rat	Recommended	Text	Supplier
Cell Type	Compatible cell types. e.g., iPSC, Mesenchymal Stem Cell, T Cell	Recommended	Text	Supplier

4.8 Biospecimens

2 additional Required fields (Specimen Type and Species). Human and animal biological samples.

Field	Definition / Picklist Values	Priority	Format	Data Source
Specimen Type	Values: Blood, Serum, Tissue, DNA, RNA, PBMCs, Bone Marrow Cells, Cord Blood Cells, Primary Cells, Other	Required	Picklist	Supplier
Species	Values: Human, Mouse, Rat, Non-Human Primate, Other	Required	Picklist	Supplier
Disease State	e.g., Normal, Melanoma, Type 2 Diabetes	Recommended	Text	Supplier

4.9 Animal Models

2 additional Required fields (Species and Strain). May require quoting/RFQ flow rather than standard cart.

Field	Definition / Picklist Values	Priority	Format	Data Source
Species	Values: Mouse, Rat, Zebrafish, Guinea Pig, Hamster, Non-Human Primate, Other	Required	Picklist	Supplier
Strain	Genetic lineage. e.g., C57BL/6, BALB/c, Sprague Dawley	Required	Text	Supplier

Genetic Modification	Values: Wild-type, Knockout, Transgenic, Knock-in, Humanized, CRISPR-Modified, Other	Recommended	Picklist	Supplier
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4.10 Equipment, Instruments & Parts

3 additional Required fields (Equipment Type, Sub-Type, Product Role). Lab instruments, automated systems, accessories, and replacement parts.

Field	Definition / Picklist Values	Priority	Format	Data Source
Equipment Type	e.g., Centrifuge, Flow Cytometer, Pipettor, Microscope, Cell Separator	Required	Text	Supplier
Sub-Type	Values: Benchtop, Floor, Handheld, Portable, Software	Required	Picklist	Supplier
Product Role	Values: Instrument, Accessory, Replacement Part, Consumable	Required	Picklist	Supplier

4.11 Controls, Calibrators & Standards

2 additional Required fields (Sub-Type and Analyte/Parameter). Quality control materials, calibrators, and reference standards.

Field	Definition / Picklist Values	Priority	Format	Data Source
Sub-Type	Values: QC Control, Calibrator, Reference Standard	Required	Picklist	Supplier
Analyte / Parameter	What the control measures. e.g., Hematopoietic Progenitor Cells, CFU	Required	Text	Supplier
Matrix	e.g., Human blood, Synthetic, Latex particle	Recommended	Text	Supplier
Regulatory Status	Values: FDA Cleared, CE Marked, RUO, IVD	Recommended	Picklist	Supplier

5. Service Listings

Services use a simplified schema compared to products. Only 6 fields are required — enough to create a searchable listing. Services without a fixed price operate under a Request for Quote (RFQ) flow, so pricing is recommended but not required.

Service categories are drawn from the SE service ontology (27 L2 categories, 236 L3 categories) rather than the 11 product categories. There are no category-specific attributes for services.

Field	Definition / Examples	Priority	Format	Data Source
Service Identification				

Unique ID	System identifier. Populated from Supplier SKU.	Required	Text	System
UNSPSC Code	8-digit UNSPSC code for the service from the full commodity taxonomy.	Required	Text	Supplier / SE
Service Name	Descriptive name as shown in search. Preserved as-is from supplier.	Required	Text	Supplier
Service Description	Extended description of the service, scope, deliverables.	Required	Text	Supplier
Category	SE service category from the service ontology (27 L2 categories, 236 L3 categories). e.g., Biology > Assay Development, Chemistry and Materials > Analytical Chemistry Services.	Required	Picklist	SE Assigned
Type	"Service" for all service listings.	Required	Picklist	SE Assigned
Pricing				
Price	Price for the service. If left blank, the listing operates under a Request for Quote (RFQ) flow.	Recommended	Numeric	Supplier
Currency	Currency code (e.g., USD, EUR, GBP). Preserved from supplier if provided; otherwise system-assigned.	Recommended	Text	Supplier / System
Unit of Measure	How the service is scoped. See Section 18 for full Service UOM picklist (17 values). e.g., Analysis, Study, Project, Day, Sample.	Recommended	Picklist	Supplier
Logistics				
Lead Time (Days)	Estimated business days to delivery or start.	Recommended	Numeric	Supplier
Ship From	Region service is delivered from. System-populated on upload.	System	Picklist	System
Site	Supplier site identifier. System-populated on upload.	System	Text	System
Remit To	Payment remittance entity. System-populated on upload.	System	Text	System
Documentation				
TDS URL	Link to service specification or scope document.	Recommended	URL	Supplier
SDS URL	Link to Safety Data Sheet (if applicable).	Recommended	URL	Supplier
COA URL	Link to Certificate of Analysis or completion.	Recommended	URL	Supplier

Additional Docs URL	Link to other supporting documentation.	Recommended	URL	Supplier
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6. Data Enrichment Strategy

Suppliers provide core required fields. Science Exchange enriches remaining attributes from public databases and product name parsing where possible. Enrichment applies to products only — services are not enriched.

Category	Primary Source	Lookup Key	Fields Enrichable
Chemicals & Solvents	PubChem	CAS Number	Molecular Formula, Weight, SMILES, InChI Key, MDL Number, PubChem CID, Hazard Classification
Molecular Biology	NCBI Gene, UniProt	Gene Symbol + Species	Gene full name, Entrez Gene ID, species validation
Proteins & Peptides	UniProt, NCBI Protein	Protein Name + Species	Molecular weight, UniProt ID, sequence length
Antibodies	Product name parsing	Product Name patterns	Target, Conjugate, Host Species, Clonality, Reactivity, Clone ID
Lab Supplies	Product name parsing	Product Name patterns	Sub-Type, Material, Sterility, Pack Size

6.1 Enrichment Process

Step 1: Supplier uploads catalog with all Required fields completed and as many Recommended fields as feasible.

Step 2: Ingestion pipeline validates required fields and flags missing data.

Step 3: For Chemicals, system queries PubChem API using CAS Number to auto-populate missing molecular identifiers.

Step 4: For Mol Bio and Protein products, gene/protein names validated against NCBI Gene and UniProt.

Step 5: For Antibodies and Lab Supplies, product names are parsed for structured attributes (Target, Conjugate, Host Species, Material, Sterility, etc.).

Step 6: Enriched data merged with supplier data. Supplier-provided values always take precedence.

7. Data Normalization Requirements

Fields requiring controlled vocabularies for consistent cross-supplier filtering. Supplier values are mapped to standard terms during ingestion.

Field	Standard Values	Normalization Rule
Unit of Measure	37 Product values + 17 Service values (see Section 18)	Map supplier abbreviations to full words. EA→Each, PK→Pack, BT→Bottle, mg→Milligrams (mg). Default to "Each" if no match.
Category	11 Product categories + service ontology	Controlled picklist. Determines which category-specific fields apply.

Physical State	Solid, Liquid, Gas, Powder, Crystal, Solution, Gel, Paste	"Powdered" → "Powder", "Fluid" → "Liquid"
Sterility	Sterile, Non-Sterile, Sterilizable	"Aseptic" → "Sterile", "Not Sterile" → "Non-Sterile"
Hazard Class.	Flammable, Corrosive, Toxic, Oxidizer, Irritant, Environmental Hazard, Health Hazard, Explosive, Non-Hazardous	Map to GHS categories. Multi-select (semicolon separated).
Storage Conditions	Room Temperature, 2-8°C, -20°C, -80°C, Desiccated, Protected from Light	Normalize temperature formats. "RT" → "Room Temperature".
Species	Human, Mouse, Rat, Rabbit, Drosophila, Zebrafish, E. coli, etc.	NCBI Taxonomy common names. "Murine" → "Mouse".

8. Unit of Measure Picklists

Separate UOM picklists for products and services. If a supplier's UOM doesn't match any value in the applicable list, default to "Each" for products or "Project" for services.

8.1 Product UOM (37 values)

Each	Pack	Kit	Bottle
Tube	Case	Bag	Plate
Set	Pair	Bundle	Slide
Unit	Gallon	Liter	Grams (g)
Milligrams (mg)	Kilograms (kg)	Milliliters	Reaction
Assay	Test	Sample	Prep
Lane	Array	Antibody	Amino Acid
Cell Line	Cell Strain	Strain	Clone
Compound	Plasmid	Protein	nmol
micromoles			

8.2 Service UOM (17 values)

Analysis	Study	Report	Project
Course	Day	hours	Months
Minute	Per session	Run	Phase
Order	Task	Setup	Donor
Animal			

8.3 UOM Normalization Rules

When suppliers provide abbreviations or alternate forms, normalize to the preferred value from the picklist above:

EA, ea → **Each** | PK, Pk → **Pack** | BT, Bt → **Bottle** | CS → **Case** | KT → **Kit** | BG → **Bag**

mg → **Milligrams (mg)** | g → **Grams (g)** | kg → **Kilograms (kg)** | mL, ml → **Milliliters** | L → **Liter**

No picklist match: BX (Box), RL (Roll), DR (Drum), VL (Vial), AM (Ampoule) → default to "Each"

9. Upload File Format

The catalog upload file must conform to these specifications for successful system ingestion.

Property	Value
File Format	CSV (UTF-8 encoded). Ready for direct system upload with no further editing.
Column Prefix	Most columns require "Spec - " prefix (note spaces around hyphen). 11 fields are exempt: Unique ID, Product Name, Product Description, Category, Price per Unit, Unit of Measure, Lead Time (Days), Ship From, Site, Type, Currency.
Empty Columns	Drop any column with zero non-empty values across the catalog.
Category Columns	Only include category-specific columns for categories present in the catalog. Shared field names (Sub-Type, Application, etc.) use a single column; the system routes by the row's Category value.
System Fields	Ship From, Site, Remit To: always leave blank (system fills on upload). Currency: preserve from supplier if provided, otherwise leave blank.
Pricing	Products: Never modify Price per Unit. Services: If Price is blank, listing operates under RFQ flow.